

Abstract

Contrastive Learning is widely used to learn shared representations between paired data modalities, such as images and text. Given paired observations X and Y , the goal is to learn embeddings $g_X(X)$ and $g_Y(Y)$ so that matched pairs are close in latent space, while mismatched pairs are separated. Although this objective is intuitive, it is not always clear what structure the learned embeddings actually capture, especially when the encoders are highly nonlinear.

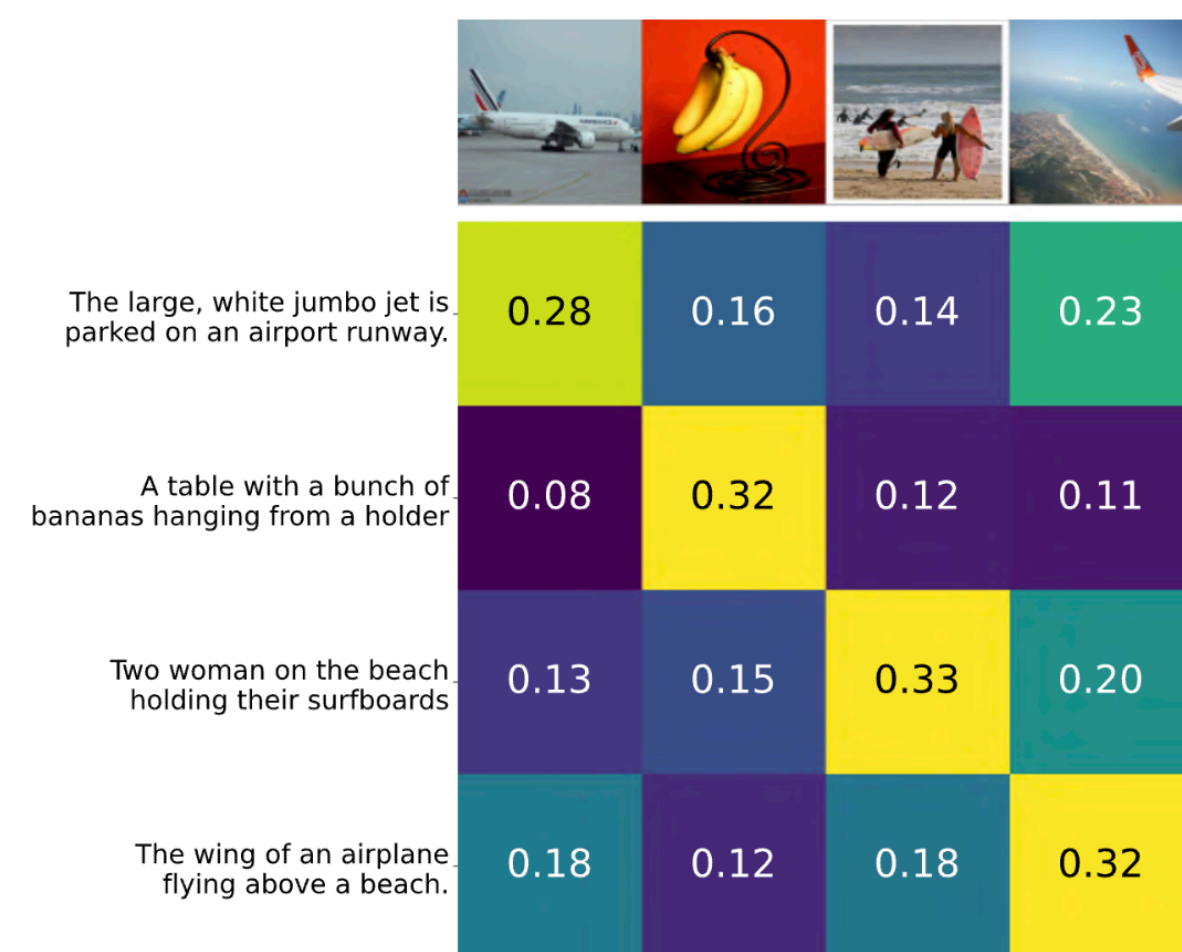


Figure 1. Contrastive learning aligns paired observations by increasing similarity between matched embeddings and reducing similarity between mismatched embeddings.

Our project studies this problem from two complementary directions. First, we analyze the linear case through CCA, where the contrastive objective has a clearer mathematical interpretation as an orthogonal projection. Our goal is to optimize

$$\max_{G,H} \mathbb{E} \left[\frac{\langle GX, HY \rangle}{\|GX\| \|HY\|} \right] \iff \min_{G,H} \text{tr}(GH^T \Sigma_{XY}) - \frac{1}{2} \text{tr}(H^T G \Sigma_{XX} G^T H \Sigma_{YY}).$$

Second, we use experiments with neural-networks to understand the behaviour of the procedure as we move beyond linear encoders.

Why Tapering?

The sample covariance is very noisy in high-dimensions. This can make canonical correlations look artificially strong, even when the two datasets are unrelated.

Tapering regularizes the covariance matrix by shrinking distant entries. This is useful when nearby variables are expected to be more related than far-away variables.

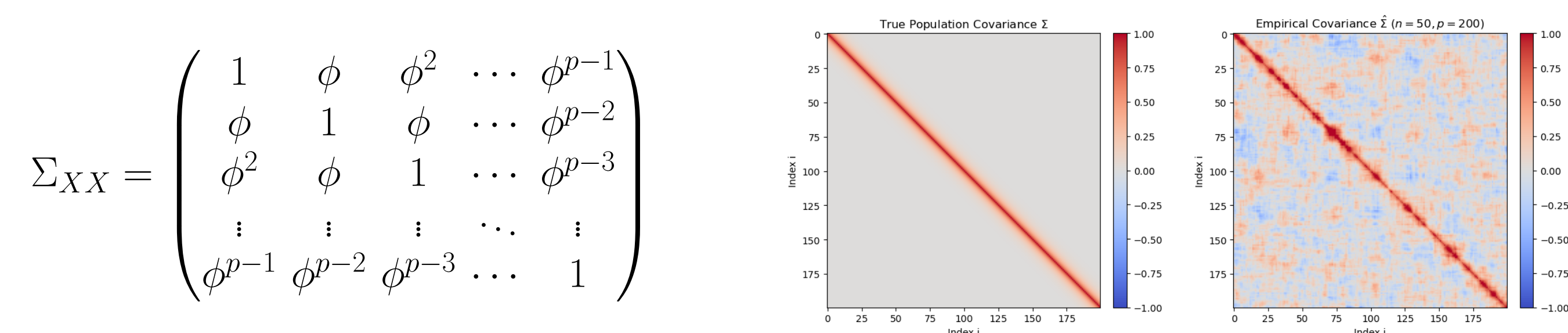


Figure 2. Expression of the covariance matrix of an **AR(1)** process and heatmaps of the population and sample covariance matrices with $p = 200$ variables and $n = 50$ samples.

- **Reduce noise** in long-range covariance estimates.
- **Stabilize CCA** by limiting inflated sample correlations.

Tapering gives us a controlled way to study when CCA is detecting real shared structure versus artifacts from noisy covariance estimates.

When the tapering penalty is applied to all three constituent matrices: $\hat{\Sigma}_{xx}$, $\hat{\Sigma}_{yy}$, and the cross-covariance $\hat{\Sigma}_{xy}$. While tapering the marginal covariance matrices $\hat{\Sigma}_{xx}$ and $\hat{\Sigma}_{yy}$ is conceptually sound, applying the same distance-based penalty to $\hat{\Sigma}_{xy}$ is not ideal. However, leaving $\hat{\Sigma}_{xy}$ intact while tapering only the marginal covariances creates an imbalance in the CCA objective, making the formulation difficult to handle mathematically.

Soft Tapering

Assume $X, Y \sim \mathcal{N}(0, K)$, where K is AR(1) covariance matrix of parameter ϕ . We define a soft tapering function as $T_{ij} = e^{-\alpha|i-j|^\beta}$ for some decay rate $\alpha > 0$. Consequently, as the distance $|i - j|$ grows, the multiplier smoothly approaches zero. Because the covariance matrix Σ follows an AR(1) structure, applying this exponential function will get us

$$(T \odot \Sigma)_{ij} = \phi^{|i-j|} e^{-\alpha|i-j|^\beta}.$$

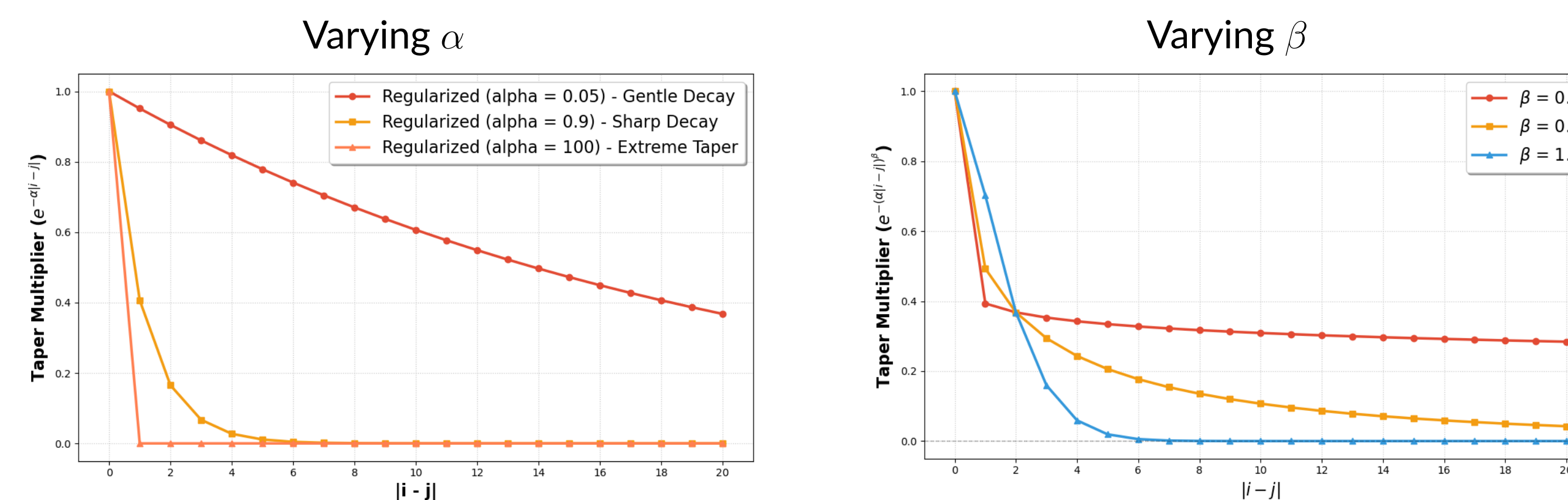


Figure 3. Exponential tapering weights under different parameter choices.

Hard Tapering

To reduce noise in sample covariance estimation, one may apply hard tapering with bandwidth b , which keeps only covariance entries i, j within distance $|i - j| \leq b$ of the diagonal and sets all others to zero. This improves locality but can destroy positive definiteness, leading to negative eigenvalues and unstable CCA, where tapered canonical correlations may become inflated far beyond 1. We explored the connection between the parameters b and ϕ on the stability of tapered CCA.

To analyze this, consider the Fourier representation of a Toeplitz AR(1) spectral density. The original and tapered matrices have continuous *symbols* given by

$$f_\phi(\theta) = 1 + 2 \sum_{k=1}^{\infty} \phi^k \cos(k\theta), \quad f_{b,\phi}(\theta) = 1 + 2 \sum_{k=1}^b \phi^k \cos(k\theta).$$

By Szegő's limit theorem, the minimum of the spectral density is asymptotically equal to the minimum of this Fourier representation, which are key to the inverse matrices in the expression for CCA.

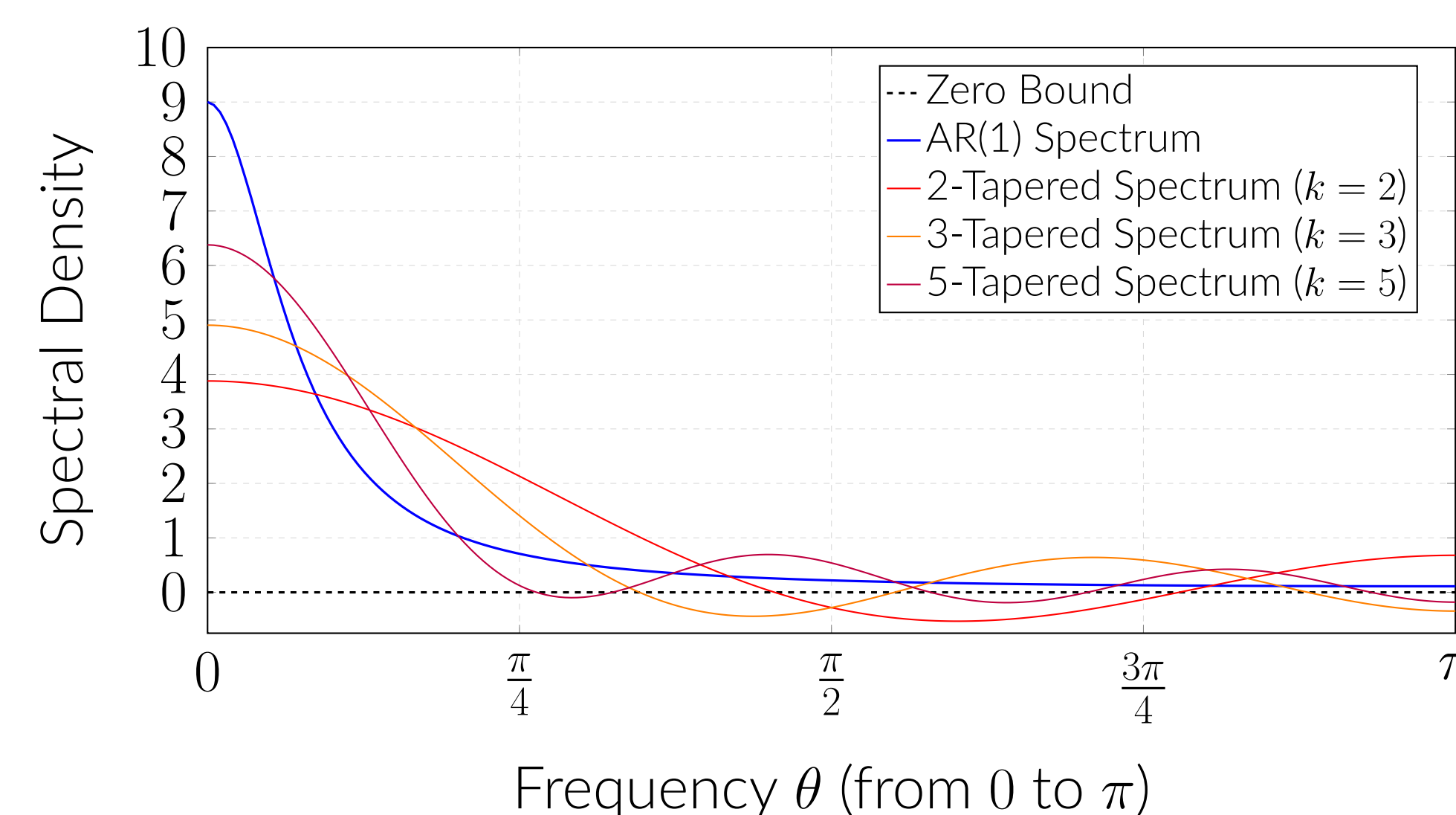


Figure 4. The asymptotic continuous spectrum of an AR(1) covariance with $\phi = 0.8$ versus its tapered versions ($k = 2, 3, 5$). As the taper lag k increases, the truncated series oscillates more rapidly. Higher k improves the approximation at lower frequencies.

Hence instability occurs precisely when $\min_\theta f_{b,\phi}(\theta) < 0$. Thus, valid tapering regimes are those where $f_{b,\phi}(\theta)$ remains nonnegative, ensuring stable CCA behavior. If we restrict values of b and ϕ to not violate the above condition, we get less correlation inflation compared to untapered CCA.

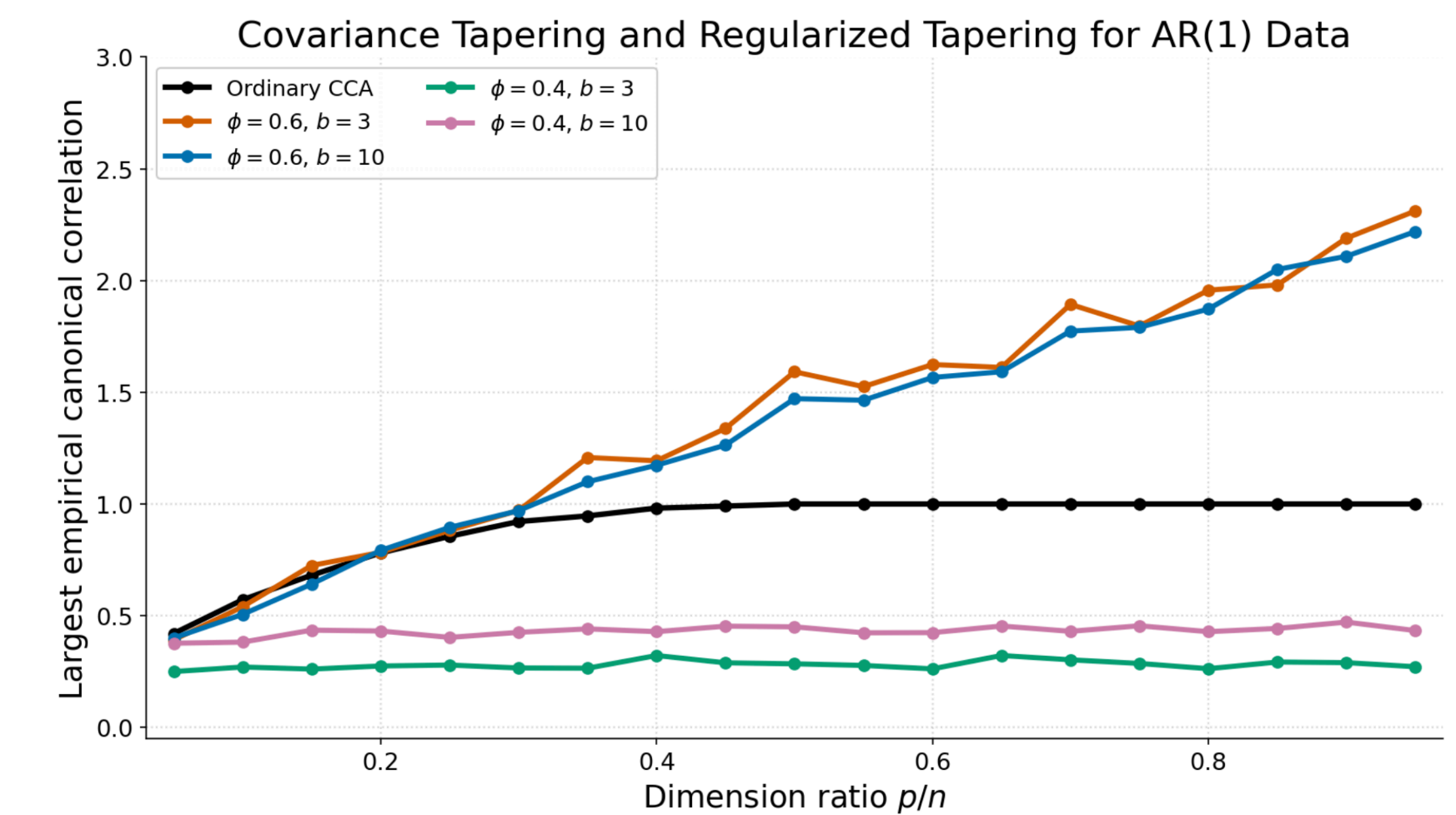


Figure 5. Ordinary CCA compared to tapered CCA with different choices of b and ϕ .

Beyond Linear CCA

CCA captures shared **linear** relationships between two modalities; here we test whether a contrastive encoder can recover nonlinear shared structure. We train paired encoders on nonlinear scalar relationships, then freeze the model and use ridge probes to inspect what each branch makes readable. The linear branch mostly captures broad trends, while the nonlinear branch and combined embedding recover curved target shapes.

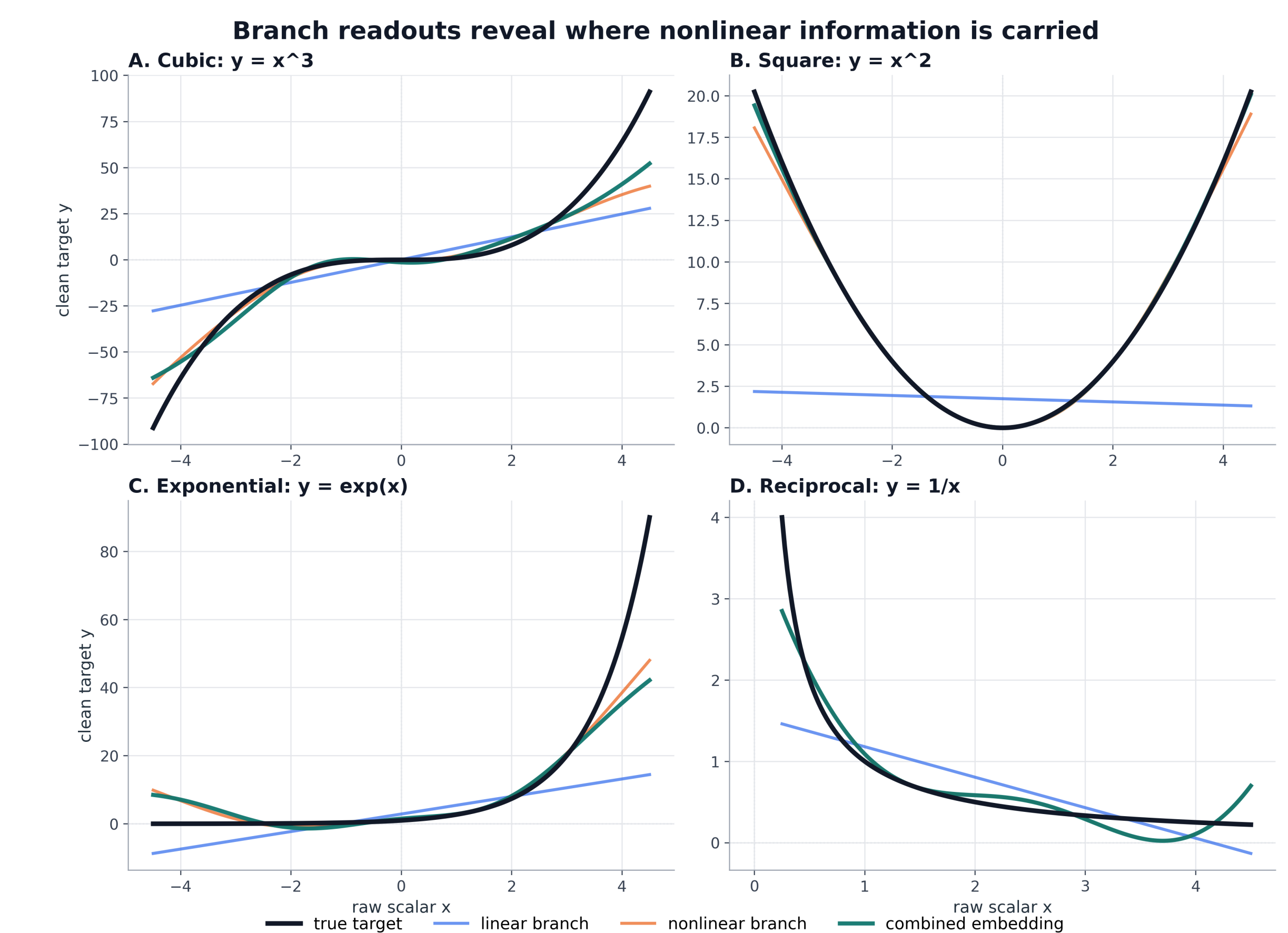


Figure 6. Branch readouts across four nonlinear targets.

Takeaway. Nonlinear contrastive encoders capture shared structure that a linear branch alone cannot recover.

References

- [1] R. Baptista, A. M. Stuart, and S. Tran, "A mathematical perspective on contrastive learning," *arXiv preprint arXiv:2505.24134*, 2025.
- [2] P. J. Bickel and E. Levina, "Covariance regularization by thresholding," *The Annals of Statistics*, vol. 36, 2008.
- [3] A. Bykhovskaya and V. Gorin, "High-dimensional canonical correlation analysis," *arXiv*, 2023.